

UD2: EJERCICIO DE VIRTUAL HOST

1. Vamos a configurar el Ubuntu Server de nuestro Virtual Box para que tenga dos interfaces de red: Uno en modo Host-only Networking (adaptador solo anfitrión) y otro como Red NAT (configurada en las preferencias, red 10.0.2.0/24 por ejemplo).

The screenshot shows the VirtualBox Network Manager interface. On the left, there are two running virtual machines: 'UbuntuDesktop' and 'UbuntuServerApache'. The main panel displays the 'Host-only Networks' tab, which contains a table of network adapters.

Nombre	IPv4 Prefix	IPv6 Prefix	Servidor DHCP
VirtualBox Host-Only Ethernet Adapter #2	192.168.141.1/24		Habilitado
VirtualBox Host-Only Ethernet Adapter	192.168.1.4/24		Inhabilitado

Below the table, the 'Adaptador' tab is selected for the 'VirtualBox Host-Only Ethernet Adapter #2'. The configuration shows the option 'Configurar adaptador manualmente' selected. The IPv4 Address is set to 192.168.141.1, the Mask of IPv4 is 255.255.255.0, the IPv6 Address is fe80::5fb5:f094:d0bf:3bbe, and the Length of IPv6 mask is 64. The 'Aplicar' button is visible.

Below this, the 'Servidor DHCP' tab is selected. The option 'Habilitar servidor' is checked. The configuration fields are as follows:

- Dirección del servidor: 192.168.141.2
- Máscara del servidor: 255.255.255.0
- Límite inferior de direcciones: 192.168.141.3
- Límite superior de direcciones: 192.168.141.254

Nombre	IPv4 Prefix	IPv6 Prefix	Servidor DHCP
NatNetwork	10.0.2.0/24	fd17:625c:f037:2::/64	Habilitado

General Options Reenvío de puertos

Nombre:

IPv4 Prefix:

☒ Enable DHCP

☐ Enable IPv6

IPv6 Prefix:


UbuntuServerApache
 Corriendo

Sistema

Memoria base: 2048 MB
 Orden de arranque: Disquete, Óptica, Disco duro
 Aceleración: Paginación anidada, Paravirtualización KVM

Pantalla

Memoria de vídeo: 16 MB
 Controlador gráfico: VMSVGA
 Servidor de escritorio remoto: Inhabilitado
 Grabación: Inhabilitado

Almacenamiento

Controlador: IDE
 IDE secundario maestro: [Unidad óptica] Vacío
 Controlador: SATA
 Puerto SATA 0: UbuntuServerApache.vdi (Normal, 8,00 GB)

Audio

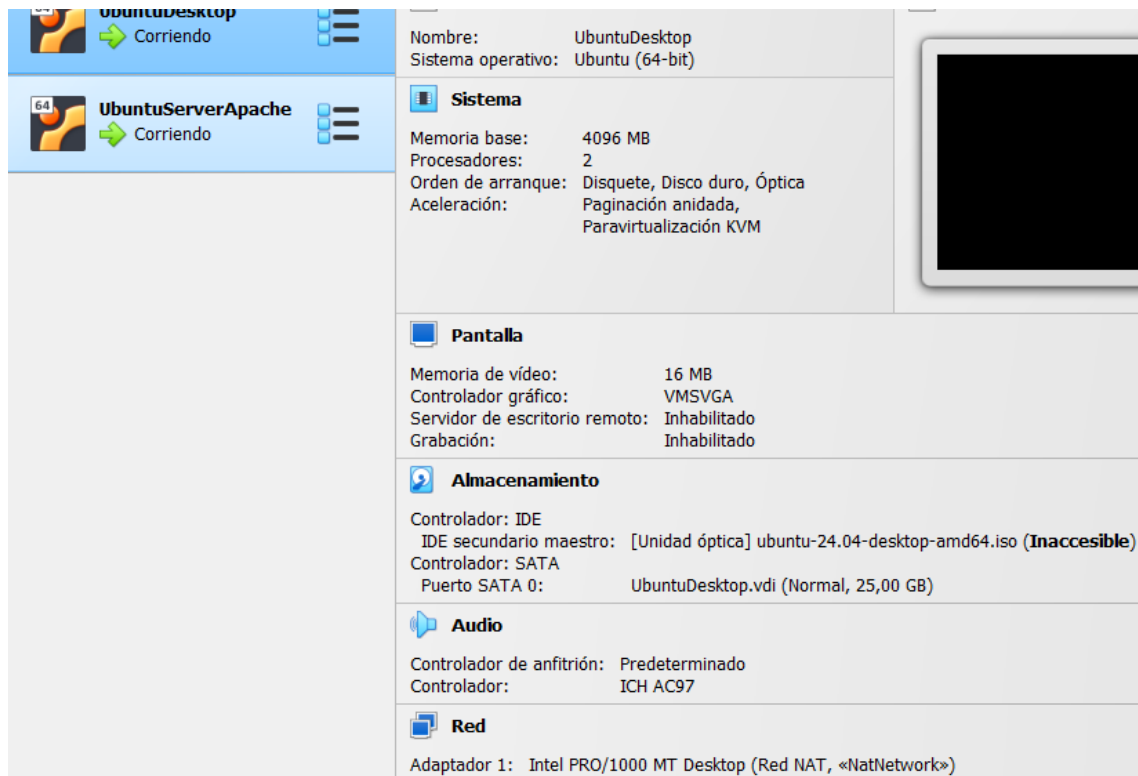
Controlador de anfitrión: Predeterminado
 Controlador: ICH AC97

Red

Adaptador 1: Intel PRO/1000 MT Desktop (Adaptador solo anfitrión, «VirtualBox Host-Only Ethernet Adapter #2»)
 Adaptador 2: Intel PRO/1000 MT Desktop (Red NAT, «NatNetwork»)



2. Instalamos un Ubuntu Desktop en otra máquina virtual que tenga solamente Red NAT.



3. Cuando esté bien configurado, vuestro Windows se podrá comunicar con el Ubuntu Server, y éste con vuestro Windows y con el Ubuntu Desktop (comprobadlo con pings)

Server

```
root@serverapache:/Lorea/comun# ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.141.3 netmask 255.255.255.0 broadcast 192.168.141.255
    inet6 fe80::a00:27ff:fe48:afdd prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:48:af:dd txqueuelen 1000 (Ethernet)
    RX packets 449 bytes 53184 (53.1 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 303 bytes 23136 (23.1 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

enp0s8: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.9 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fe80::a00:27ff:fe92:53e8 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:92:53:e8 txqueuelen 1000 (Ethernet)
    RX packets 47455 bytes 71561476 (71.5 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 12023 bytes 740537 (740.5 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 344 bytes 29966 (29.9 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 344 bytes 29966 (29.9 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

Desktop

```
root@ubuntudesktop:/home/lorea# ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.6 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fe80::a00:27ff:fe75:1f0 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:75:01:f0 txqueuelen 1000 (Ethernet)
    RX packets 240 bytes 305058 (305.0 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 164 bytes 17113 (17.1 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Bucle local)
    RX packets 188 bytes 18431 (18.4 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 188 bytes 18431 (18.4 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

Ping Desktop-Server

```
root@ubuntudesktop:/home/lorea# ping -c4 10.0.2.9
PING 10.0.2.9 (10.0.2.9) 56(84) bytes of data.
64 bytes from 10.0.2.9: icmp_seq=1 ttl=64 time=0.870 ms
64 bytes from 10.0.2.9: icmp_seq=2 ttl=64 time=0.599 ms
64 bytes from 10.0.2.9: icmp_seq=3 ttl=64 time=0.785 ms
64 bytes from 10.0.2.9: icmp_seq=4 ttl=64 time=0.690 ms

--- 10.0.2.9 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3064ms
rtt min/avg/max/mdev = 0.599/0.736/0.870/0.101 ms
root@ubuntudesktop:/home/lorea# nano /etc/hosts
```

Ping Server-Desktop

```
root@serverapache:/Lorea/comun# ping -c4 10.0.2.6
PING 10.0.2.6 (10.0.2.6) 56(84) bytes of data.
64 bytes from 10.0.2.6: icmp_seq=1 ttl=64 time=0.606 ms
64 bytes from 10.0.2.6: icmp_seq=2 ttl=64 time=0.491 ms
64 bytes from 10.0.2.6: icmp_seq=3 ttl=64 time=0.968 ms
64 bytes from 10.0.2.6: icmp_seq=4 ttl=64 time=1.21 ms

--- 10.0.2.6 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3076ms
rtt min/avg/max/mdev = 0.491/0.819/1.213/0.287 ms
root@serverapache:/Lorea/comun#
```

Ping Windows-Server

```
C:\Users\lorea>ping 192.168.141.3

Haciendo ping a 192.168.141.3 con 32 bytes de datos:
Respuesta desde 192.168.141.3: bytes=32 tiempo<1m TTL=64
Respuesta desde 192.168.141.3: bytes=32 tiempo<1m TTL=64
Respuesta desde 192.168.141.3: bytes=32 tiempo<1m TTL=64
Respuesta desde 192.168.141.3: bytes=32 tiempo<1m TTL=64

Estadísticas de ping para 192.168.141.3:
    Paquetes: enviados = 4, recibidos = 4, perdidos = 0
        (0% perdidos),
    Tiempos aproximados de ida y vuelta en milisegundos:
        Mínimo = 0ms, Máximo = 0ms, Media = 0ms
```

Ping Server-Windows

```
rtt min/avg/max/mdev = 0.491/0.819/1.213/0.287 ms
root@serverapache:/Lorea/comun# ping -c4 192.168.14.33
PING 192.168.14.33 (192.168.14.33) 56(84) bytes of data.
64 bytes from 192.168.14.33: icmp_seq=1 ttl=127 time=1.74 ms
64 bytes from 192.168.14.33: icmp_seq=2 ttl=127 time=1.66 ms
64 bytes from 192.168.14.33: icmp_seq=3 ttl=127 time=1.58 ms
64 bytes from 192.168.14.33: icmp_seq=4 ttl=127 time=1.43 ms

--- 192.168.14.33 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 1.432/1.602/1.738/0.113 ms
```

4. En el servidor web el contenido de cada virtual host irá en una carpeta dentro del directorio **/"vuestroNombre"**. Mirad un ejemplo en apache2.conf de cómo hacerlo.

```
mkdir /Lorea
mkdir /Lorea/comun
mkdir /Lorea/interna
mkdir /Lorea/externa
```

5. Tendremos 3 virtual host: interna, externa, común. El contenido de las mismas es irrelevante, pero tienen que diferenciarse para saber a cuál estamos accediendo.

```
nano /Lorea/comun/index.html
```

```
GNU nano 7.2
<html>
<h1>Pagina comun</h1>
</html>
```

`nano /Lorea/interna/index.html`

```
GNU nano 7.2
<html>
<h1>Pagina interna</h1>
</html>
```

`nano /Lorea/externa/index.html`

```
GNU nano 7.2
<html>
<h1>Pagina externa</h1>
</html>
```

6. **interna** solamente será accesible desde la interfaz de la red interna en el puerto 80, es decir, desde el Ubuntu Desktop. **externa** solamente será accesible desde vuestro Windows y el puerto 80. **común** será accesible desde ambas máquinas, pero desde el puerto 8080 solamente

`nano /etc/apache2/sites-available/comun.conf`

```
GNU nano 7.2 /etc/apache2/sites-available/comun.conf
<VirtualHost *:8080>
    # The ServerName directive sets the request scheme, hostname and port that
    # the server uses to identify itself. This is used when creating
    # redirection URLs. In the context of virtual hosts, the ServerName
    # specifies what hostname must appear in the request's Host: header to
    # match this virtual host. For the default virtual host (this file) this
    # value is not decisive as it is used as a last resort host regardless.
    # However, you must set it for any further virtual host explicitly.
    ServerName paginaweb.org

    ServerAdmin webmaster@localhost
    DocumentRoot /Lorea/comun

    # Available loglevels: trace8, ..., trace1, debug, info, notice, warn,
    # error, crit, alert, emerg.
    # It is also possible to configure the loglevel for particular
    # modules, e.g.
    #LogLevel info ssl:warn

    ErrorLog ${APACHE_LOG_DIR}/error.log
    CustomLog ${APACHE_LOG_DIR}/access.log combined

    # For most configuration files from conf-available/, which are
    # enabled or disabled at a global level, it is possible to
    # include a line for only one particular virtual host. For example the
    # following line enables the CGI configuration for this host only
    # after it has been globally disabled with "a2disconf".
    #Include conf-available/serve-cgi-bin.conf
</VirtualHost>
```

nano /etc/apache2/sites-available/interna.conf

```
GNU nano 7.2 /etc/apache2/sites-available/in
<VirtualHost 10.0.2.9:5500>
# The ServerName directive sets the request scheme, hostname and port that
# the server uses to identify itself. This is used when creating
# redirection URLs. In the context of virtual hosts, the ServerName
# specifies what hostname must appear in the request's Host: header to
# match this virtual host. For the default virtual host (this file) this
# value is not decisive as it is used as a last resort host regardless.
# However, you must set it for any further virtual host explicitly.
ServerName paginaweb.org

ServerAdmin webmaster@localhost
DocumentRoot /Lorea/interna

# Available loglevels: trace8, ..., trace1, debug, info, notice, warn,
# error, crit, alert, emerg.
# It is also possible to configure the loglevel for particular
# modules, e.g.
#LogLevel info ssl:warn

ErrorLog ${APACHE_LOG_DIR}/error.log
CustomLog ${APACHE_LOG_DIR}/access.log combined

# For most configuration files from conf-available/, which are
# enabled or disabled at a global level, it is possible to
# include a line for only one particular virtual host. For example the
# following line enables the CGI configuration for this host only
# after it has been globally disabled with "a2disconf".
#Include conf-available/serve-cgi-bin.conf
</VirtualHost>
```

nano /etc/apache2/sites-available/externa.conf

```
GNU nano 7.2 /etc/apache2/sites-availabl
<VirtualHost 192.168.141.3:80>
# The ServerName directive sets the request scheme, hostname and port that
# the server uses to identify itself. This is used when creating
# redirection URLs. In the context of virtual hosts, the ServerName
# specifies what hostname must appear in the request's Host: header to
# match this virtual host. For the default virtual host (this file) this
# value is not decisive as it is used as a last resort host regardless.
# However, you must set it for any further virtual host explicitly.
ServerName paginaweb.org

ServerAdmin webmaster@localhost
DocumentRoot /Lorea/externa

# Available loglevels: trace8, ..., trace1, debug, info, notice, warn,
# error, crit, alert, emerg.
# It is also possible to configure the loglevel for particular
# modules, e.g.
#LogLevel info ssl:warn

ErrorLog ${APACHE_LOG_DIR}/error.log
CustomLog ${APACHE_LOG_DIR}/access.log combined

# For most configuration files from conf-available/, which are
# enabled or disabled at a global level, it is possible to
# include a line for only one particular virtual host. For example the
# following line enables the CGI configuration for this host only
# after it has been globally disabled with "a2disconf".
#Include conf-available/serve-cgi-bin.conf
</VirtualHost>
```

Habilitar los hosts virtuales

`cd /etc/apache2/sites-available`

`a2ensite comun`

`a2ensite interna`

`a2ensite externa`

Deshabilitar el default

`a2dissite 000-default.conf`

Modificar el archivo de configuración de apache2

`nano /etc/apache2/apache2.conf`

```
<Directory />
    Options FollowSymLinks
    AllowOverride None
    Require all denied
</Directory>

<Directory /usr/share>
    AllowOverride None
    Require all granted
</Directory>

<Directory /var/www/>
    Options Indexes FollowSymLinks
    AllowOverride None
    Require all granted
</Directory>

<Directory /Lorea/>
    Options Indexes FollowSymLinks
    AllowOverride None
    Require all granted
</Directory>

#<Directory /Lorea/comun>
#     Require ip 192.168.141
#</Directory>
```

Modificar el archivo de configuración de ports

`nano /etc/apache2/ports.conf`

```
#Listen 80
Listen 192.168.141.3:80
Listen 10.0.2.9:5500
Listen 8080

<IfModule ssl_module>
    Listen 443
</IfModule>

<IfModule mod_gnutls.c>
    Listen 443
</IfModule>
```

Reiniciar Apache2

`systemctl restart apache2`

7. A las páginas (a todas) se accederá escribiendo en el navegador **paginaweb.org** (y el puerto si fuera necesario).

Desktop-etik frogak:

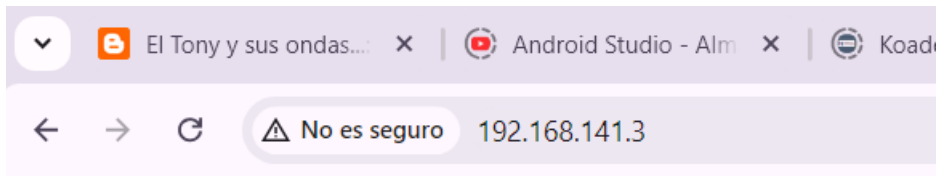
Pagina Comun



Pagina Interna



Desde Windows



Pagina externa

www.paginaweb.org???

8. Prueba la funcionalidad de la opción **Indexes** para el virtual host **común**.

¿¿¿Si en el servidor desaparece el archivo de index.html q pasaría???Para realizar esta prueba:

`cd /Lorea/comun`

Crear dos fitxeros cualquiera

`nano Fitxategi.txt`

`nano Fitxategi2.txt`

`ls`

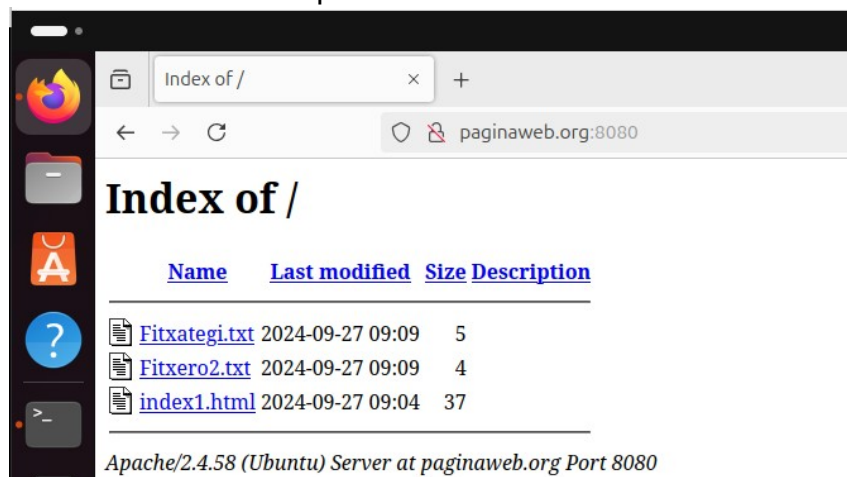
```
root@serverapache:/etc/apache2/sites-available#  
root@serverapache:/Lorea/comun# ls  
Fitxategi.txt Fitxero2.txt index.html  
root@serverapache:/Lorea/comun#
```

Cambiar index.html por index1.html

```
mv: cannot stat 'index1.html': No such file or directory  
root@serverapache:/Lorea/comun# mv index.html index1.html  
root@serverapache:/Lorea/comun# ls  
Fitxategi.txt Fitxero2.txt index1.html  
root@serverapache:/Lorea/comun#
```

`Systemctl restart apache2`

Prueba desde Desktop:



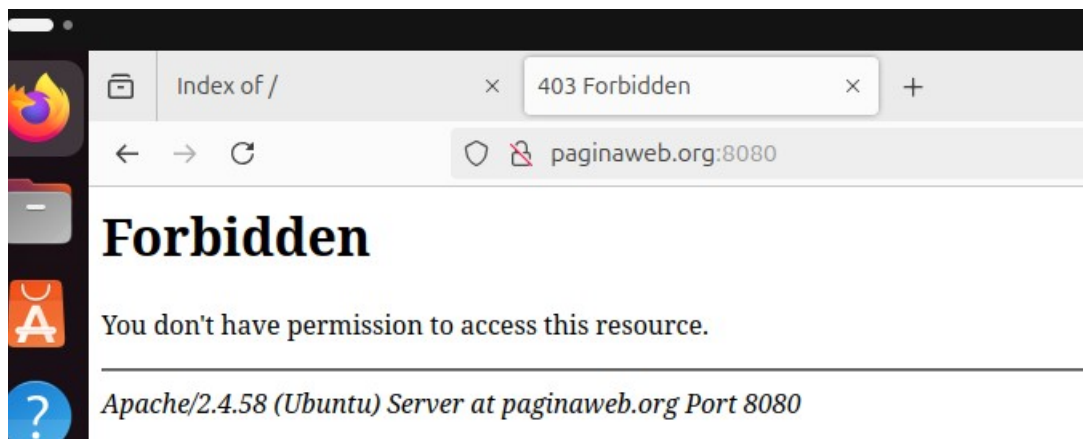
9. Prueba la funcionalidad de la directiva **Require ip** en el virtual host común para permitir el acceso solamente desde el Ubuntu Desktop.

`nano /etc/apache2/apache2.conf`

```
Directory /Lorea/>
    Options Indexes FollowSymLinks
    AllowOverride None
    Require all granted
/Directory>

Directory /Lorea/comun>
    Require ip 192.168.141
/Directory>
```

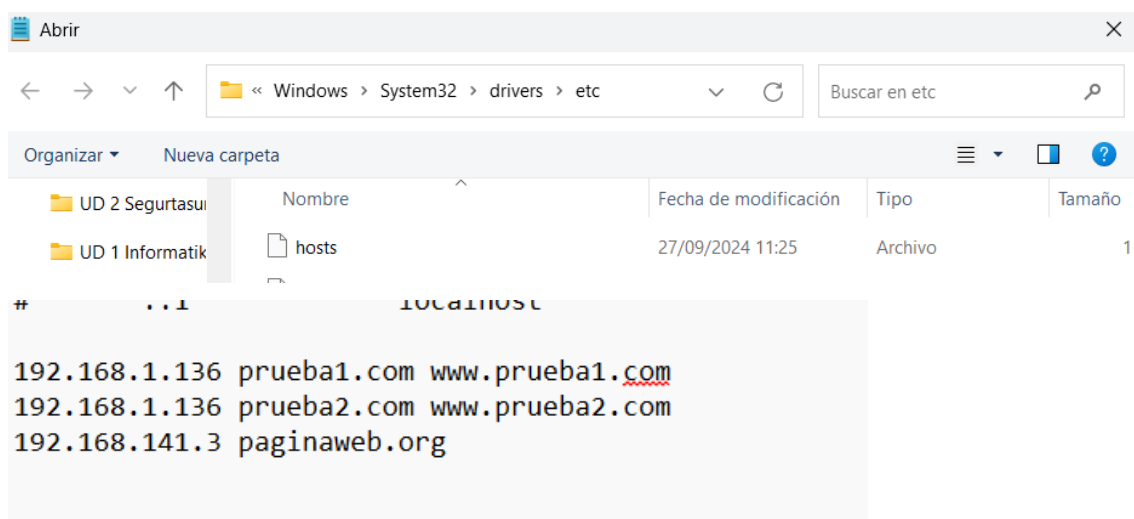
`Systemctl restart apache2`
`Systemctl status apache2`



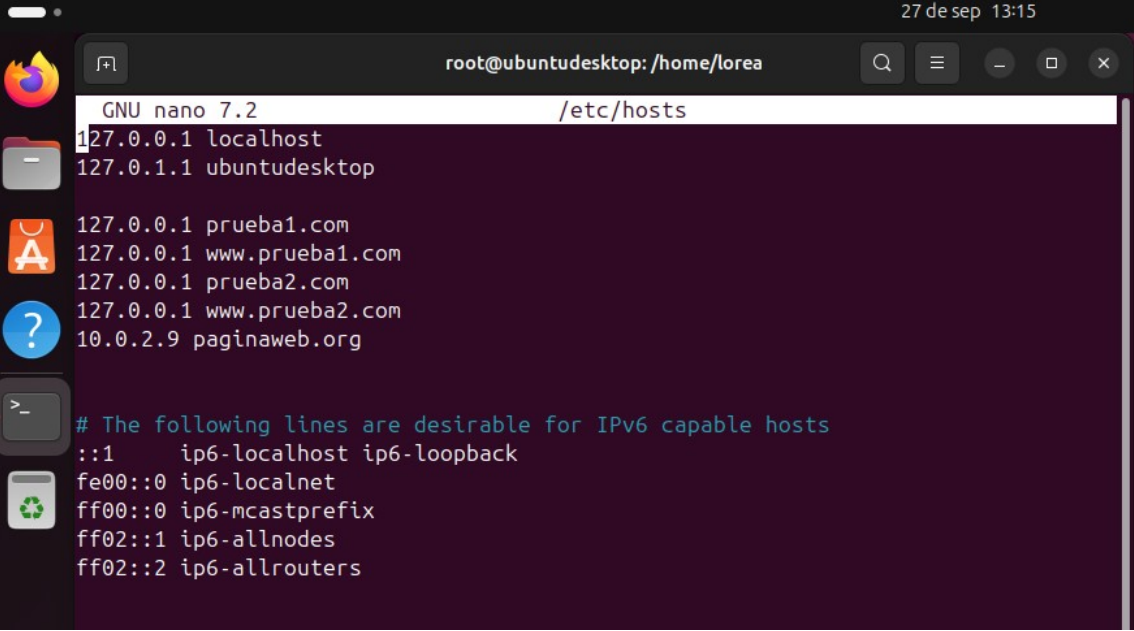
OHARRA:

Gogoratu bai Desktop eta Windows-en hosts-en aldaketa egitea:

Windows-en (gogoratu administratzaile bezala sartzea)



Desktop



The screenshot shows a desktop environment with a terminal window open. The terminal window has a title bar that reads "root@ubuntudesktop: /home/lorea". The terminal is running the GNU nano 7.2 text editor, editing the file /etc/hosts. The content of the file is as follows:

```
127.0.0.1 localhost
127.0.1.1 ubuntudesktop

127.0.0.1 prueba1.com
127.0.0.1 www.prueba1.com
127.0.0.1 prueba2.com
127.0.0.1 www.prueba2.com
10.0.2.9 paginaweb.org

# The following lines are desirable for IPv6 capable hosts
::1      ip6-localhost ip6-loopback
fe00::0  ip6-localnet
ff00::0  ip6-mcastprefix
ff02::1  ip6-allnodes
ff02::2  ip6-allrouters
```

The desktop environment includes a sidebar with icons for the Dash (fire), Home (folder), Applications (Ubuntu logo), Dash (?), and Recycle Bin. The top right corner of the window shows the date and time: "27 de sep 13:15".